

ACN Manual Solution

Practical No. 1:

a) Identify IP allocations and Internet Service Providers for a student network Using WHOIS.

b) Set up IP addresses and subnet masks on given network devices

Conclusion : Successfully executed the following :-

a) Identify IP allocations and Internet Service Providers for a student network Using WHOIS.

b) Set up IP addresses and subnet masks on given network devices

Practical related questions

1. Which command is commonly used to query WHOIS information for a domain or IP address?

Answer :

The command commonly used to query WHOIS information for a domain or IP address is : **whois**

Example:

whois example.com

For an IP address:

whois 8.8.8.8

2. What information can you typically obtain from a WHOIS query for an IP address?

Answer :

A WHOIS query for an IP address can typically provide:

- IP address and address range
- Internet Service Provider (ISP) or organization that owns the IP
- Organization name and contact details
- Country or geographic region
- Autonomous System Number (ASN)
- Network registration information
- Registration and allocation dates
- Abuse/contact information

3. Which subnet mask corresponds to a network with 256 possible IP addresses?

Answer :

A network with **256 possible IP addresses** requires **8 host bits** because:

$$2^8 = 256$$

So the prefix is **/24**.

Subnet Mask: 255.255.255.0 (CIDR: /24)

4. What is the purpose of assigning a subnet mask to a network device?

Answer :

The purpose of assigning a **subnet mask** to a network device is to determine which part of the IP address represents the **network** and which part represents the **host**.

It helps the device:

- Identify whether another device is on the **same network**.
- Determine where to send network traffic.
- Divide a large network into **smaller subnets**.
- Use IP addresses efficiently.

Example: 255.255.255.0 indicates that the first 24 bits identify the network, while the remaining 8 bits identify individual hosts.

5. If a network device is assigned the IP address \192.168.1.10 \ with a subnet mask of \255.255.255.0 \, what is the network address?

Answer :

Given:

- **IP Address:** 192.168.1.10
- **Subnet Mask:** 255.255.255.0 (/24)

The first three octets represent the network, and the last octet represents the host.

Therefore:

Network Address = 192.168.1.0

Practical No. 2: Identify and resolve network issues using network diagnostic tools like ping, tracer, show, debug commands.

Conclusion : Successfully identified and resolved network issues using network diagnostic tools like ping, tracer, show, debug commands.

Practical related questions

1. What is the primary purpose of the 'ping' command?

Answer :

The primary purpose of the **ping command** is to **check whether a device is reachable over a network** and measure the **time taken for packets to travel to the destination and back**.

2. Which protocol does the 'tracer' command primarily rely on?

Answer :

The **tracer** command primarily relies on the **ICMP (Internet Control Message Protocol)**.

- **Windows:** tracer uses ICMP Echo Request messages.
- **Linux/macOS:** The equivalent traceroute commonly uses UDP or ICMP, depending on the implementation/options.

3. What does the command 'show ip route' display?

Answer :

The **show ip route** command displays the **IP routing table** of a router.

It typically shows:

- Destination networks
- Next-hop addresses
- Routing protocols/routes
- Network interfaces used to reach destinations
- Metrics and administrative distances

4. Which command would you use to monitor real-time packet flow on a Cisco router?

Answer :

The command used to monitor **real-time packet flow** on a Cisco router is:

debug ip packet

It displays information about IP packets as they are processed by the router.

Practical No. 3: Run a Network Communication Script on "Kali Linux"

Conclusion : Successfully executed a Network Communication Script on "Kali Linux"

Practical related questions

1. Which Python library is commonly used for socket programming to establish network communication?

Answer :

The Python library commonly used for socket programming and network communication is the **socket** library.

Example:

```
import socket
```

It provides functions for creating **TCP and UDP connections**, sending and receiving data, and communicating between network devices.

2. What command in Kali Linux is used to capture and analyze network packets in real-time?

Answer :

The command commonly used in **Kali Linux** to capture and analyze network packets in real time is:

```
tcpdump
```

Example:

```
sudo tcpdump
```

It captures packets from a network interface and displays their details in the terminal.

3. Which tool in Kali Linux is primarily used for network scanning and discovering open ports?

Answer :

The tool in **Kali Linux** primarily used for network scanning and discovering open ports is **Nmap (Network Mapper)**.

Example:

```
nmap 192.168.1.1
```

It can be used to:

- Discover active hosts
- Identify **open ports**
- Detect running services
- Gather basic information about network devices

4. What is the purpose of the '-sS' flag in the Nmap command?

Answer :

The **-sS** flag in Nmap performs a **TCP SYN scan**.

It sends SYN packets to target ports to determine whether they are open, closed, or filtered, without completing the full TCP connection.

Example:

```
nmap -sS 192.168.1.1
```

5. Which command in Kali Linux can be used to test connectivity to a specific IP address or hostname?

Answer :

The command used in **Kali Linux** to test connectivity to a specific IP address or hostname is **ping**.

Example:

```
ping 192.168.1.1
```

Or:

```
ping google.com
```

It sends ICMP packets to the target and checks whether it responds.

Practical No. 4: *Configure basic routing protocols using any relevant software/virtual lab.

Conclusion : Successfully configured basic routing protocols using any relevant software/virtual lab.

Practical related questions

1. You set up a RIP routing configuration between two routers, but one router is not receiving routes. What are the first three troubleshooting steps you take?

Answer :

The first three troubleshooting steps for a **RIP routing problem** are:

1. **Check the interface status and IP addresses**
2. show ip interface brief

Make sure the interfaces are **up/up** and have correct IP addresses.

3. **Verify the RIP configuration**
4. show running-config

Check that RIP is enabled, the correct network statements are configured, and both routers use the same RIP version if required.

5. **Check the routing table and RIP updates**
6. show ip route
7. show ip protocols

Confirm whether RIP routes are being learned and whether RIP is sending/receiving updates.

2. In OSPF, what happens if two routers have different area IDs but belong to the same subnet? How would you resolve this issue?

Answer :

If two OSPF routers are on the **same subnet but have different Area IDs**, they will **not form an OSPF neighbour adjacency**. As a result, they will not exchange OSPF routing information over that link.

How to resolve it:

1. Check the OSPF configuration on both routers.
2. Ensure the interfaces connecting the routers are assigned to the **same OSPF area**.
3. Use commands such as:
4. show ip ospf interface
5. show ip ospf neighbor
6. Correct the area configuration if necessary, for example:
7. router ospf 1
8. network 192.168.1.0 0.0.0.255 area 0

3. After configuring EIGRP, you notice high latency in packet forwarding. What parameters would you check and modify to optimize performance? What is difference between == , equals() & compareTo()?

Answer :

After configuring **EIGRP**, if you notice high latency, check and optimize these parameters:

1. **Bandwidth** – Verify that the configured bandwidth accurately reflects the available link capacity.
2. show interfaces
3. **Delay** – Check the interface delay, as EIGRP uses delay as an important metric.
4. show interfaces
5. **EIGRP Metric** – Check the routing metric and ensure appropriate **K-values** are configured on all EIGRP routers.
6. show ip protocols
7. **Load and congestion** – Check whether interfaces are overloaded or experiencing packet loss.
8. **Routing table and topology** – Verify that EIGRP is selecting the best available path.
9. show ip route
10. show ip eigrp topology
11. **Modify bandwidth/delay if necessary** to accurately represent the network path and encourage EIGRP to select a better route.

Difference between ==, equals(), and compareTo() in Java

Method	Purpose	Example	Result
==	Compares primitive values or object references	a == b	true / false
equals()	Compares object contents/logical equality	a.equals(b)	true / false
compareTo()	Compares objects to determine their ordering	a.compareTo(b)	Negative, 0, or positive

Practical No. 5: Capture and Analyze ICMPv4 Packets using appropriate tool

Conclusion : Successfully captured and analyzed ICMPv4 Packets using appropriate tool

Practical related questions

1. Which command in tcpdump captures only ICMPv4 packets?

Answer :

The tcpdump command used to capture **only ICMPv4 packets** is:

tcpdump icmp

For a specific interface:

sudo tcpdump -i eth0 icmp

2. What steps should you follow to filter ICMPv4 traffic in Wireshark?

Answer :

To filter **ICMPv4 traffic in Wireshark**, follow these steps:

1. **Open Wireshark** and select the network interface you want to monitor.
2. Start capturing packets.
3. In the **Display Filter** bar, enter:
4. icmp
5. Press **Enter** or click the filter/apply button.
6. Wireshark will display **only ICMPv4 packets**, such as ping request and reply packets.

3. How can you differentiate between an Echo Request and an Echo Reply in Wireshark?

Answer :

In Wireshark, you can differentiate between **ICMP Echo Request** and **ICMP Echo Reply** by checking the **ICMP Type** field.

- **Echo Request: ICMP Type 8** — sent by the source to test connectivity.
- **Echo Reply: ICMP Type 0** — response sent back by the destination.

In Wireshark:

1. Apply the filter:
2. icmp
3. Select an ICMP packet.
4. Expand **Internet Control Message Protocol (ICMP)** in the packet details.
5. Check the **Type** field.

4. What information does an ICMP Destination Unreachable packet provide?

Answer :

An **ICMP Destination Unreachable** packet indicates that a packet **could not reach its intended destination**.

It provides information about **why the destination is unreachable**, such as:

- **Network unreachable** – the destination network cannot be reached.
- **Host unreachable** – the specific host cannot be reached.
- **Port unreachable** – the destination port is not available.
- **Protocol unreachable** – the required protocol is unavailable.
- **Fragmentation needed** – the packet needs fragmentation but cannot be fragmented.

5. Why is the Time Exceeded message useful in network troubleshooting?

Answer :

The **ICMP Time Exceeded** message is useful in network troubleshooting because it indicates that a packet's **TTL (Time To Live) has reached zero** before reaching its destination.

It helps to:

- Identify **routing loops** in a network.
 - Determine where packets are being **delayed or stopped**.
 - Trace the path packets take through a network.
 - Diagnose routing problems using tools such as **traceroute** or **tracert**.
-

Practical No. 6: *Configure, diagnose and troubleshoot TCP and UDP connection issues using diagnostic tools like netstat, wireshark, iperf

Conclusion : Successfully configured, diagnosed and troubleshooted TCP and UDP connection issues using diagnostic tools like netstat, wireshark, iperf

Practical related questions

1. How can you use netstat to identify active TCP connections on a server?

Answer :

You can use the **netstat** command with the -t and -n options to identify active TCP connections:

netstat -tn

- **-t** → Displays TCP connections.
- **-n** → Displays IP addresses and port numbers in numerical form.

To show only **active/established TCP connections**:

netstat -tn | grep ESTABLISHED

2. What does the TIME_WAIT state in netstat indicate about a TCP connection?

Answer :

The **TIME_WAIT** state in netstat indicates that a TCP connection has been **closed, but the system is temporarily keeping the connection information**.

It is used to:

- Ensure delayed packets from the old connection do not interfere with a new connection.
- Allow the final TCP packets to be properly acknowledged.
- Prevent connection-related errors when the same IP/port pair is reused.

Example:

TCP 192.168.1.10:5000 192.168.1.20:80 TIME_WAIT

3. How would you use Wireshark to detect TCP retransmissions in a network capture?

Answer :

To detect **TCP retransmissions** in Wireshark:

1. **Open the captured .pcap file** in Wireshark.
2. In the **Display Filter** bar, enter:
3. tcp.analysis.retransmission
4. Press **Enter**.
5. Wireshark will display packets identified as **TCP retransmissions**.
6. Select a packet and check the **Info** column and TCP details to examine the retransmitted sequence number.

Common related filters:

tcp.analysis.fast_retransmission

Shows fast retransmissions.

tcp.analysis.lost_segment

Helps identify possible missing TCP segments.

4. What steps would you take in Wireshark to analyze a failed TCP three-way handshake?**Answer :**

To analyze a **failed TCP three-way handshake** in Wireshark:

1. **Start or open a packet capture** containing the failed connection attempt.
2. Apply the filter:
3. tcp
4. Look for the normal three-way handshake:
 - **SYN** → Client sends a connection request.
 - **SYN-ACK** → Server responds.
 - **ACK** → Client confirms the connection.
5. Check whether the **SYN-ACK or ACK is missing**.
6. Look for **TCP retransmissions**, using:
7. tcp.analysis.retransmission
8. Check for **RST (Reset)** packets, which may indicate that the connection was refused or forcibly closed.
9. Examine the **source/destination IP addresses and port numbers** to identify which device is causing the failure.

5. Write a program to convert Integer object value into primitive datatype byte, short and double value**Answer :**

```
public class IntegerConversion {
    public static void main(String [] args) {
        Integer num = 100;
        byte b = num.byteValue();
        short s = num.shortValue();
        double d = num.doubleValue();

        System.out.println("Integer value: " + num);
        System.out.println("Byte value: " + b);
        System.out.println("Short value: " + s);
        System.out.println("Double value: " + d);
    }
}
```

Practical No. 7: Configure DNS using relevant software.

Conclusion : Successfully configured DNS using relevant software.

Practical related questions

1. Explain the purpose of DNS in a computer network and how it helps users access websites.

Answer :

Purpose of DNS in a Computer Network

DNS (Domain Name System) is used to translate **human-readable domain names** into **IP addresses** that computers use to communicate with servers.

For example, when a user enters:

www.google.com

DNS finds the corresponding IP address of Google's server. The browser then uses that IP address to connect to the website.

How DNS helps users:

1. User enters a **domain name** in the browser.
2. The DNS resolver searches for the domain's **IP address**.
3. The IP address is returned to the user's device.
4. The browser connects to the web server using that IP address.
5. The website is displayed to the user.

2. Describe the basic steps involved in the DNS name resolution process.

Answer :

1. **User enters a domain name**
For example, www.example.com is entered in a web browser.
2. **DNS cache is checked**
The device checks its local DNS cache to see if the IP address is already known.
3. **DNS resolver is contacted**
If the address is not cached, the request is sent to a **DNS resolver**, usually provided by the ISP or network administrator.
4. **DNS servers are queried**
The resolver contacts the appropriate DNS servers, starting with the **root DNS server**, then the **TLD server** (such as .com), and finally the **authoritative DNS server**.
5. **IP address is returned**
The authoritative server provides the IP address for the requested domain.
6. **Connection is established**
The DNS resolver sends the IP address back to the user's device, and the browser uses it to connect to the web server.

3. How would you configure a DNS server to resolve a domain name to its corresponding IP address?

Answer :

1. **Install a DNS server** such as **BIND (Berkeley Internet Name Domain)** on the server.
2. **Create a DNS zone** for the domain, for example:
3. example.com
4. **Create an A record** in the zone file that maps the domain name to the IP address:
5. example.com. IN A 192.168.1.10
6. **Configure the DNS server** to load the new zone and verify that the configuration has no errors.
7. **Restart or reload the DNS service** so the changes take effect.
8. **Test the configuration** using a command such as:
9. nslookup example.com

If configured correctly, the DNS server will return 192.168.1.10 when example.com is queried.

4. What troubleshooting steps would you take if a DNS server is not resolving domain names correctly?

Answer :

1. **Check network connectivity**
2. ping <DNS-server-IP>

Make sure the client can reach the DNS server.

3. **Check DNS server configuration**
Verify that the DNS service is running and the configuration files contain the correct **zone and record information**.
 4. **Check DNS records**
Make sure the required **A, AAAA, CNAME, or other records** are correctly configured.
 5. **Test DNS resolution**
Use:
 6. nslookup example.com or dig example.com
 7. **Check the client's DNS settings**
Verify that the client is using the correct DNS server address.
 8. **Check firewall and port access**
Ensure **DNS port 53** is not blocked. DNS commonly uses both **UDP and TCP**.
 9. **Clear DNS cache**
Clear stale DNS information on the client or DNS server and try the query again.
 10. **Check DNS logs**
Examine the DNS server logs for configuration errors, failed queries, or other problems.
-

Practical No. 8: Configure FTP using relevant software.

Conclusion : Successfully configured FTP using relevant software.

Practical related questions

1. What steps are involved in setting up an FTP server on a computer?

Answer :

1. **Install FTP server software**
Install an FTP server such as **vsftpd** on Linux or **IIS FTP Server** on Windows.
2. **Configure the FTP server**
Set the server's IP address, FTP port (**21**), user accounts, and access permissions.
3. **Create FTP users and directories**
Create users and specify the directories they are allowed to access.
4. **Set file and folder permissions**
Configure appropriate **read/write permissions** for users.
5. **Configure the firewall**
Allow FTP traffic through **TCP port 21** and configure passive-mode ports if required.
6. **Start the FTP service**
Start and enable the FTP server service.
7. **Test the connection**
Connect using an FTP client such as FileZilla or the command line:
8. `ftp <server-ip>`

2. How would you create and manage user accounts for secure access to an FTP server?

Answer :

1. **Create a user account** on the server with a unique username and strong password.
2. **Assign the user a specific FTP directory** so they can access only the required files.
3. **Set file and directory permissions** according to the user's role, such as read-only or read/write.
4. **Restrict users to their home directory** using a feature such as chroot to prevent unauthorized access to other directories.
5. **Disable anonymous FTP access** if it is not required.
6. **Regularly review and manage accounts** by changing passwords, removing unused accounts, and updating permissions.
7. For stronger security, use **SFTP/FTPS** instead of plain FTP because standard FTP sends credentials and data without encryption.

For example, on a Linux server:

```
sudo adduser ftpuser
```

Then assign the appropriate directory and permissions to ftpuser.

3. Which FTP command is used to upload a file to the server, and which is used to download a file? Explain with an example.

Answer :

In FTP:

- **put** → Used to **upload** a file from the local computer to the FTP server.
- **get** → Used to **download** a file from the FTP server to the local computer.

1. Upload a File – put

```
ftp> put report.txt
```

This uploads **report.txt** from the local computer to the current directory on the FTP server.

2. Download a File – get

```
ftp> get report.txt
```

This downloads **report.txt** from the FTP server to the local computer.

Example

```
ftp> put assignment.pdf
```

```
local: assignment.pdf remote: assignment.pdf
```

```
200 PORT command successful.
```

```
226 Transfer complete.
```

```
ftp> get notes.txt
```

```
local: notes.txt remote: notes.txt
```

```
200 PORT command successful.
```

```
226 Transfer complete.
```

4. After connecting to an FTP server, how would you list the files in a directory and change to another directory?

Answer :

After connecting to an FTP server:

1. **List files in the current directory** using the ls or dir command:
2. ftp> ls

This displays the files and directories in the current location.

3. **Change to another directory** using the cd command:
4. ftp> cd documents

This changes the current directory to **documents**.

5. **Go back to the parent directory:**
6. ftp> cd ..

Example

```
ftp> ls  
documents  
images  
report.pdf
```

```
ftp> cd documents  
ftp> ls  
assignment.pdf  
notes.txt
```

Practical No. 9: Monitor Network Traffic using Browser Developer tools.

Conclusion : Successfully Monitored Network Traffic using Browser Developer tools.

Practical related questions

1. How do you open the network tab in a browser's developer tools?

Answer :

1. Open your web browser (Chrome, Firefox, or Edge).
2. Press **F12** or **Ctrl + Shift + I** to open Developer Tools.
3. Click the **Network** tab.
4. Reload the webpage using **Ctrl + R** to capture the network activity.

The Network tab displays **HTTP/HTTPS requests, response status codes, request methods, loading times, headers, and transferred data.**

2. How do you check if a website's image or file has loaded successfully using network tab?

Answer :

1. Open **Developer Tools** using F12 or Ctrl + Shift + I.
2. Select the **Network** tab.
3. Reload the webpage using Ctrl + R.
4. Find the required **image or file** in the list of network requests. You can use the **Img, Media, or Fetch/XHR** filters if available.
5. Check the **Status** column:
 - **200 OK** → The file loaded successfully.
 - **404 Not Found** → The file could not be found.
 - **403 Forbidden** → Access to the file is denied.
 - **500** → Server-side error.
6. Click the request to view its **Headers, Response, and Timing** details.

3. How can you find out how much time a web page takes to load?

Answer :

1. Open **Developer Tools** using F12 or Ctrl + Shift + I.
2. Select the **Network** tab.
3. Reload the webpage using Ctrl + R.
4. Look at the **Network activity and timing information.**
5. The browser shows the time taken for individual resources such as HTML, CSS, JavaScript, and images.
6. You can also check the **Load** or **Finish** time at the bottom of the Network panel to see the overall loading time.

4. How can you tell if a file on a webpage failed to load?

Answer :

1. Open **Developer Tools** using F12 or Ctrl + Shift + I.
2. Select the **Network** tab.
3. Reload the webpage using Ctrl + R.
4. Find the required file, such as an **image, CSS, JavaScript, or font file**.
5. Check the **Status** column:
 - **200 OK** → File loaded successfully.
 - **404 Not Found** → File does not exist or the URL is incorrect.
 - **403 Forbidden** → Access is denied.
 - **500 Internal Server Error** → Server-side problem.
6. Failed requests may also appear **highlighted in red** in the Network panel.

5. How do you check the size of a file that loads on a website?

Answer :

1. Open **Developer Tools** using F12 or Ctrl + Shift + I.
 2. Select the **Network** tab.
 3. Reload the webpage using Ctrl + R.
 4. Find the file you want to check, such as an **image, CSS, JavaScript, or PDF**.
 5. Look at the **Size** column to see how much data was transferred.
 6. For more details, click the file and check **Headers → Content-Length** or the **Size** information in the request details.
-

Practical No. 10: Design a simple network for SDN using Mininet.

Conclusion : Successfully designed a simple network for SDN using Mininet.

Practical related questions

1. What command will you use to verify that Mininet is correctly installed on your system, and what is the expected output indicating a successful installation?

Answer :

Use the following command in the terminal:

```
sudo mn --test pingall
```

This starts a basic Mininet topology and tests connectivity between the hosts.

If Mininet is installed correctly, you should see output similar to:

```
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2
*** Adding switches:
s1
*** Adding links:
...
*** Ping: testing ping reachability
h1 -> h2
h2 -> h1
*** Results: 0% dropped (2/2 received)
*** Stopping 2 hosts
*** Stopping 1 switches
*** Stopping 1 controllers
*** Done
```

The important indication of successful installation is:

***** Results: 0% dropped**

2. Create a simple network topology with one switch and two hosts and run the pingall command within the Mininet CLI. What does the output tell you about the connectivity between the hosts?

Answer :

To create a topology with **one switch and two hosts**, use:

```
sudo mn --topo single,2
```

This creates:

```
h1 ---- s1 ---- h2
```

Then, inside the **Mininet CLI**, run:

```
mininet> pingall
```

Expected Output

```
*** Ping: testing ping reachability
```

```
h1 -> h2
```

```
h2 -> h1
```

```
*** Results: 0% dropped (2/2 received)
```

Explanation

The pingall command tests connectivity between **all hosts** in the Mininet network.

- **0% dropped** → Both hosts can communicate successfully.
- **100% dropped** → The hosts cannot communicate.
- Any percentage between 0% and 100% → Some connectivity problems exist.

3. What steps do you take to enter the host's CLI? Execute a ping command. Describe the expected results and what they indicate about the network connectivity?

Answer :

Assuming the topology has two hosts (h1 and h2) connected to a switch:

1. Enter the Mininet CLI

Start Mininet:

```
sudo mn --topo single,2
```

2. Enter the CLI of Host h1

From the Mininet prompt, use:

```
mininet> xterm h1
```

This opens a terminal for **h1**.

Alternatively, you can execute a command directly from the Mininet CLI:

```
mininet> h1 ping -c 4 h2
```

3. Execute the ping command

Inside the h1 terminal:

```
ping -c 4 10.0.0.2
```

Here, 10.0.0.2 is typically the IP address of h2.

4. Expected Result

You should see output similar to:

64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=0.1 ms

64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.1 ms

64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.1 ms

64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.1 ms

4 packets transmitted, 4 received, 0% packet loss

What does this indicate?

- **4 packets received / 0% packet loss** → h1 and h2 have successful network connectivity.
- **100% packet loss** → h1 cannot reach h2, indicating a connectivity or configuration problem.
- **High response time** → There may be network delay or performance issues.

4. sudo mn --topo tree, depth=2,fanout=2 --mac --controller=default. How many hosts and switches are created in this tree topology?

Answer :

For the Mininet command:

sudo mn --topo tree, depth=2,fanout=2 --mac --controller=default

The tree topology has:

- **Depth = 2**
- **Fanout = 2**

Calculation

Switches:

- Level 1: 1 switch
- Level 2: 2 switches
- **Total switches = 3**

Hosts:

- Each of the 2 switches at the second level has 2 hosts.
 - **Total hosts = $2 \times 2 = 4$**
-

Practical No. 11: Using Ping and Latency tools to perform the following :-

- i) Measure latency and packet loss over time using any suitable tool e.g. PingPlotter.**
 - ii) Analyze network packets to detect performance bottlenecks using any suitable tool e.g. Wireshark.**
-

Conclusion : Successfully used Ping and Latency tools to perform the following :-

- i) Measure latency and packet loss over time using any suitable tool e.g. PingPlotter.
- ii) Analyze network packets to detect performance bottlenecks using any suitable tool e.g. Wireshark.

Practical related questions

1. Use PingPlotter to measure the latency to 8.8.8.8 for 5 minutes. What is the average latency and is there any packet loss? Explain what it indicates about the network connection.

Answer :

PingPlotter Test to 8.8.8.8

To perform this test:

1. Open **PingPlotter**.
2. Enter **8.8.8.8** as the target.
3. Set the sampling interval as required.
4. Run the test for **5 minutes**.
5. Check the **Avg (Average)** latency and **Packet Loss** columns.

Since the actual PingPlotter results are not provided, the exact average latency and packet loss **cannot be determined from the question alone**.

Example Interpretation

Suppose the 5-minute test shows:

- **Average latency:** 25 ms
- **Packet loss:** 0%

This indicates a **stable and responsive network connection** to 8.8.8.8.

If packet loss is present, for example **5%**, it indicates that some packets are not reaching the destination or their replies are not returning, which may indicate **network congestion, an unstable connection, or a problem along the route**.

2. Compare the latency and packet loss when testing two different websites (e.g. google.com and yahoo.com). Which one has better performance and why?

Answer :

Website	Average Latency	Packet Loss	Performance
google.com	25 ms	0%	Better
yahoo.com	45 ms	2%	Lower

Explanation

In this example, **google.com has better performance** because it has:

- **Lower average latency (25 ms)** → Faster response time.
- **0% packet loss** → All packets successfully reached the destination and returned.

Yahoo.com has higher latency and some packet loss, indicating a potentially less efficient or less stable network path.

The difference can be caused by **server location, routing paths, network congestion, peering arrangements, or server/network conditions.**

3. Identify and explain what you observe in the PingPlotter graph if there is a sudden spike in latency or appearance of red bars. What could be the possible reasons?

Answer :

In PingPlotter, if you observe a **sudden spike in latency**, the graph will show a sharp increase in response time. **Red bars** generally indicate **packet loss or a significant latency problem** at that point in the route.

Possible Reasons

1. **Network congestion** – Heavy traffic can increase latency and cause packet loss.
2. **Unstable Wi-Fi connection** – Weak signal or interference can cause dropped packets.
3. **Router or ISP problems** – A busy or malfunctioning router can introduce delays.
4. **Routing issues** – Problems on a particular network hop can cause increased latency.
5. **Server-side congestion** – The destination server may be overloaded.
6. **Temporary network interruption** – Short-term connectivity problems can produce sudden spikes.

Important: If the red bars occur only at an intermediate hop but the later hops and final destination show normal latency with no packet loss, that hop may simply be **deprioritizing ICMP responses** rather than actually dropping forwarding traffic.

4. Capture live traffic using Wireshark and apply the filter icmp. Identify at least 3 ICMP packets and explain what type of message each one is?

Answer :

1. Open **Wireshark** and select the active network interface, such as Wi-Fi or Ethernet.
2. Start capturing packets.
3. In the **Display Filter** bar, enter:
4. icmp
5. Generate some ICMP traffic by running:
6. ping 8.8.8.8
7. Stop the capture after a few seconds.

8. Examine the **Info** column and expand **Internet Control Message Protocol** in the packet details.

Example ICMP Packets

ICMP Packet	Type	Description
Echo Request	Type 8	Sent by the source device to check whether another device is reachable.
Echo Reply	Type 0	Sent back by the destination in response to an Echo Request.
Destination Unreachable	Type 3	Indicates that a packet could not reach its destination, such as when a network or host is unreachable.

For a normal ping test, you will commonly see:

ICMP Echo Request

ICMP Echo Reply

ICMP Echo Request

ICMP Echo Reply

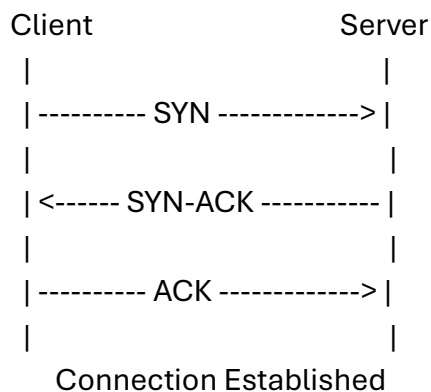
5. Analyze a captured TCP session using Wireshark. Identify the three-way handshake (SYN, SYN-ACK, ACK) and explain its purpose.

Answer :

1. **Open Wireshark** and load the captured .pcap file.
2. Apply the filter:
3. tcp
4. Find the packets at the beginning of a TCP connection.
5. Identify the following three packets:

Step	TCP Packet	Description
1	SYN	Client requests to establish a TCP connection with the server.
2	SYN-ACK	Server acknowledges the request and sends its own synchronization request.
3	ACK	Client acknowledges the server's SYN, completing the handshake.

Example



In Wireshark, you can also use:

`tcp.flags.syn == 1`

to find SYN packets and inspect the **SYN**, **ACK**, and **Sequence Number** fields.

Purpose of the Three-Way Handshake

- **Establish a reliable TCP connection.**
- Synchronize the **sequence numbers** between the client and server.
- Confirm that both devices are ready to communicate.
- Ensure that both sides can send and receive TCP segments.

6. Using the Wireshark statistics feature, find out which protocol (e.g., TCP, UDP, HTTP) is used most in your captured traffic. What does this tell you about the type of activity on the network?

Answer :

1. Open the captured file in **Wireshark**.
2. Go to **Statistics → Protocol Hierarchy**.
3. Check the number and percentage of packets for each protocol, such as **TCP, UDP, ICMP, HTTP, DNS**, etc.
4. Identify the protocol with the highest percentage of traffic.

Example

Suppose the Protocol Hierarchy shows:

Protocol	Percentage
TCP	70%
UDP	20%
ICMP	5%
HTTP	5%

In this example, **TCP is the most-used protocol**.

What does this indicate?

- **TCP is dominant** → The network is likely handling activities such as web browsing, file transfers, or other applications requiring reliable connections.
 - **UDP is dominant** → There may be activities such as DNS queries, streaming, VoIP, or online gaming.
 - **ICMP is dominant** → The capture may mainly contain network testing or troubleshooting traffic such as ping.
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Practical No. 12: Multimedia Traffic Analysis

i) Capture analyze HTTP Video streaming traffic using any suitable tool e.g. Wireshark.

ii) Monitor RTP (Real Time Transport Protocol) packets from a multimedia stream using any suitable tool e.g. Wireshark.

Conclusion : Successfully used Multimedia Traffic Analysis for the following :-

i) Capture analyze HTTP Video streaming traffic using any suitable tool e.g. Wireshark.

ii) Monitor RTP (Real Time Transport Protocol) packets from a multimedia stream using any suitable tool e.g. Wireshark.

Practical related questions

1. Using Wireshark, how can you identify the video segments being downloaded during HTTP streaming?

Answer :

1. Open **Wireshark** and start capturing traffic while playing an online video.
2. Apply the HTTP filter:
3. http

For HTTPS traffic, HTTP contents are encrypted, so you generally need appropriate decryption keys to inspect the requests and responses.

4. Look for **HTTP GET requests** made by the video player.
5. Select a GET request and inspect the **Request URI** or **File Data** fields.
6. Video segments may have filenames or extensions such as:
7. .mp4
8. .ts
9. .m4s
10. Check the **Length** or transferred data size to identify larger video segments.
11. You can use **Statistics → HTTP → Requests** or **File → Export Objects → HTTP** (for unencrypted HTTP) to examine transferred objects.

Example

You may see requests such as:

GET /video/segment1.m4s

GET /video/segment2.m4s

GET /video/segment3.m4s

These indicate that the video player is downloading the video in **small segments rather than as one complete file**.

2. What steps would you follow to capture and filter only the HTTP traffic related to a video being streamed on YouTube?

Answer :

1. **Open Wireshark** and select your active network interface, such as Wi-Fi or Ethernet.
2. Start the packet capture.
3. Open YouTube in a browser and start playing a video.
4. After a few seconds, stop the capture.
5. To view TLS traffic, use the display filter:
6. `tls`
7. If the traffic uses **HTTP/2 over TLS**, you can also try:
8. `http2`
9. For **HTTP/3/QUIC**, use:
10. `quic`
11. To focus on YouTube-related connections, inspect the **destination IP addresses, DNS queries, or TLS Server Name Indication (SNI)** for YouTube/Google video servers.
12. Examine packet sizes, timing, and data flow to observe the video streaming activity.

Important Note

A filter such as:

`http`

will usually **not show YouTube's video traffic**, because YouTube uses encrypted HTTPS-based protocols.

3. After capturing HTTP streaming traffic, how can you determine whether there were any TCP retransmissions affecting playback quality?

Answer :

1. **Open the captured file** in Wireshark.
2. Apply the filter:
3. `tcp.analysis.retransmission`
4. Wireshark will display packets identified as **TCP retransmissions**.
5. Check the **Time, Source, Destination, and Info** columns to see when and where retransmissions occurred.
6. You can also check for:
7. `tcp.analysis.fast_retransmission`
to identify fast retransmissions.
8. Compare the retransmission times with periods when the video **buffered, paused, or experienced reduced quality**.

Interpretation

- **Few or no retransmissions** → TCP delivery was generally reliable and retransmissions were unlikely to affect playback significantly.
- **Many retransmissions** → Packets were being lost and had to be resent, which can increase delay and potentially cause **buffering or playback interruptions**.
- **Repeated retransmissions during the same period as buffering** → This provides stronger evidence that packet loss contributed to the playback problem.

4. How can you apply a filter in Wireshark to view only RTP packets in a captured call or media stream?

Answer :

1. Open the captured .pcap file in **Wireshark**.
2. In the **Display Filter** bar, enter:
3. rtp
4. Press **Enter**.
5. Wireshark will display only packets that it has identified as **RTP traffic**.
6. Select an RTP packet and expand **Real-time Transport Protocol** to examine fields such as:
 - Sequence number
 - Timestamp
 - SSRC
 - Payload type

If rtp Shows No Packets

If Wireshark has not automatically recognized the traffic as RTP, you can identify the UDP stream first:

udp

Then right-click a relevant UDP packet and select:

Decode As → RTP

After decoding, apply:

rtp

5. What information in the RTP packet helps you detect packet loss or jitter?

Answer :

1. Sequence Number

- Each RTP packet has a sequence number.
- Missing or skipped sequence numbers indicate **packet loss**.
- Example: 100, 101, 103 → packet 102 may have been lost.

2. Timestamp

- Indicates the timing of the media data in the RTP stream.
- Variations in the arrival time compared with the timestamps can indicate **jitter**.

3. Arrival Time

- Wireshark records when each packet arrives.
- Irregular differences between packet arrival times indicate **network jitter**.

4. SSRC (Synchronization Source)

- Identifies the source of the RTP stream and helps Wireshark distinguish between different media streams.

In Wireshark

You can use:

Telephony → RTP → RTP Streams

and then select a stream and choose **Analyze** to view statistics such as **packet loss, jitter, sequence errors, and packet timing**.
